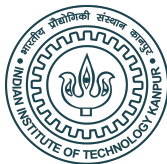


Lecture 5

Univariate Posterior Summarization

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(Edited from the lecture notes by Brian J. Reich, NC State)



Bayes' Theorem

- ▶ In Bayesian statistics, we select the prior, $\pi(\theta)$, and the likelihood, $f(y|\theta)$.
- ▶ Based on these two pieces of information, our main goal is to compute the posterior $p(\theta|y)$.
- ▶ Bayes' theorem is the mathematical formula to convert the likelihood and prior to the posterior.
- ▶ Bayes theorem:

$$p(\theta|y) = \frac{f(y|\theta)\pi(\theta)}{m(y)}.$$

- ▶ This holds for discrete (PMF) and continuous (PDF) cases.
- ▶ As in the formula for a conditional distribution, $m(y)$ is just the normalizing constant required so that $\int p(\theta|y)d\theta = 1$.

Examples of a univariate posterior

- ▶ Here, we have only a single unknown parameter, θ .

- ▶ For example, we say the model is

Likelihood: $Y|\theta \sim \text{Binomial}(N, \theta)$,

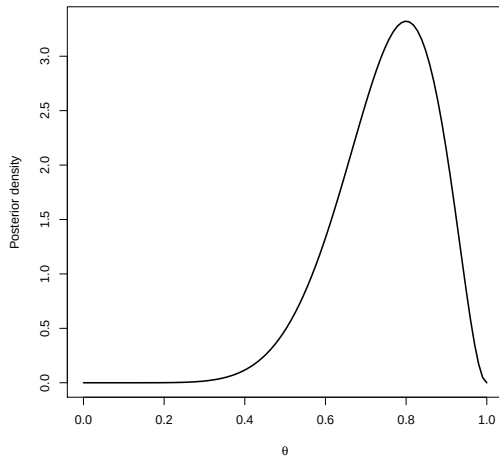
Prior: $\theta \sim \text{Beta}(a, b)$.

- ▶ We saw that the posterior is then

$$\theta|Y \sim \text{Beta}(Y + a, N - Y + b).$$

- ▶ The posterior is a *distribution* that can be plotted as on the next slide.

Summarizing a univariate posterior



In this beta/binomial example, $Y = 8$, $N = 10$ and $a = b = 1$.

Summarizing a univariate posterior

- ▶ A plot of the posterior tells the whole story.
- ▶ However, to be more concise we typically use a few numerical summaries of the distribution.
- ▶ This is particularly important when there are many parameters.
- ▶ The posterior can be summarized like any other distribution, by say the mean, variance, skewness, etc.

Point estimators

- ▶ A **point estimator** is a one number summary used to estimate the unknown parameter.
- ▶ For example, we might use the posterior mean (or median) as the “best guess” of θ .
- ▶ The posterior mean is

$$\hat{\theta} = E(\theta|Y) = \int \theta p(\theta|Y) d\theta.$$

- ▶ For the Beta/Binomial example, we have $\hat{\theta} = \frac{Y+a}{n+a+b}$.
- ▶ This is an alternative to the sample proportion $\hat{\theta} = Y/n$.

MAP estimator

- ▶ The posterior mode is called the maximum a posteriori (MAP) estimator.
- ▶ The MAP estimator is

$$\hat{\theta} = \arg \max_{\theta} p(\theta | Y) = \arg \max_{\theta} \{\log[f(Y|\theta)] + \log[\pi(\theta)]\}.$$

- ▶ If the prior is uniform (i.e., flat), the MAP is the MLE.
- ▶ The MAP is easier to compute than the posterior mean.

MAP estimator

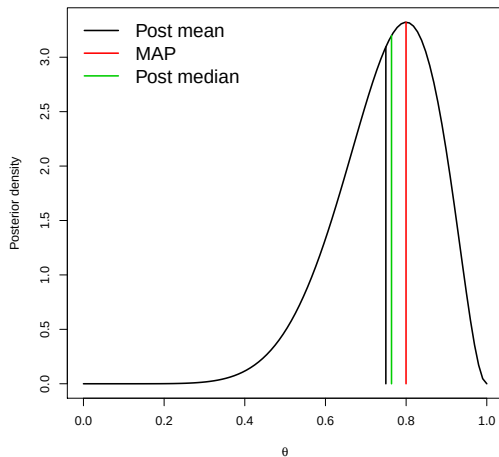
- ▶ Let $Y|\theta \sim \text{Binomial}(n, \theta)$ and $\pi(\theta) = \mathbb{1}(\theta \in [0, 1])$. Find the MAP estimator of θ .

MAP estimator

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- ▶ The MAP estimator is

$$\begin{aligned}\hat{\theta} = \arg \max_{\theta} p(\theta|Y) &= \arg \max_{\theta} \{\log[f(Y|\theta)] + \log[\pi(\theta)]\} \\ &= \arg \max_{\theta \in [0,1]} \{\log[f(Y|\theta)]\} \\ &= Y/n.\end{aligned}$$

MAP estimator

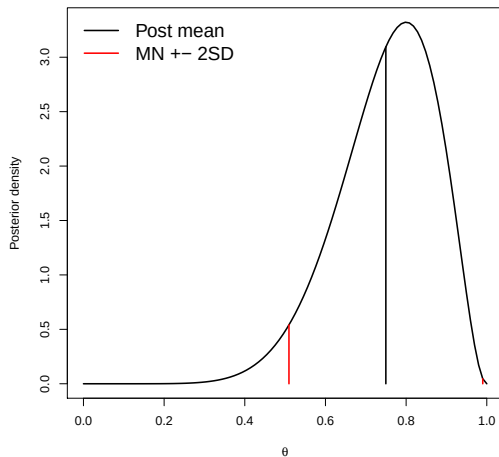


In this beta/binomial example $Y = 8$, $N = 10$ and $a = b = 1$.

Uncertainty measures

- ▶ Sometimes a point estimate is sufficient, but more often we need to quantify uncertainty.
- ▶ The **posterior standard deviation** is one measure of uncertainty.
- ▶ If the posterior is approximately normal, then the mean plus/minus two standard deviation units captures 95% of the posterior probability.
- ▶ The posterior standard deviation is analogous to but fundamentally different than the frequentist **standard error**.
- ▶ The standard error is the standard deviation of $\hat{\theta}$'s sampling distribution.

Uncertainty measures



In this beta/binomial example $Y = 8$, $N = 10$ and $a = b = 1$.

Credible intervals

- ▶ In addition to standard error, uncertainty can be quantified using a **credible interval**.

- ▶ The interval (l, u) is a $100(1 - \alpha)\%$ posterior credible interval/set if

$$\text{Prob}(l < \theta < u | Y) = 1 - \alpha.$$

- ▶ Interpretation of a 95% credible interval: “given the data and prior, I am 95% certain that θ is between l and u ”.
- ▶ This is analogous but different than a **confidence interval**.

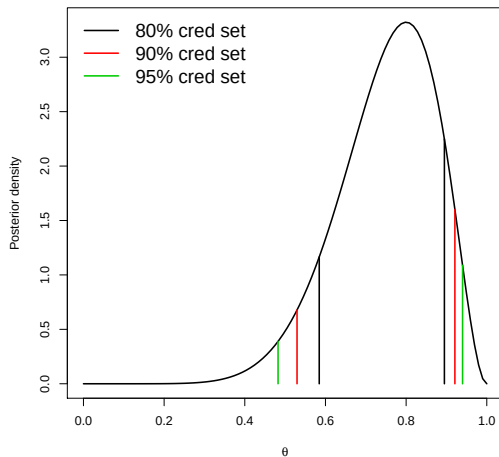
Credible intervals

- ▶ Credible sets are not unique.
- ▶ Let q_τ be the τ quantile of the posterior so that

$$\text{Prob}(\theta < q_\tau | Y) = \tau.$$

- ▶ Then $(q_{0.00}, q_{0.95})$, $(q_{0.01}, q_{0.96})$, etc. are all valid 95% credible sets.
- ▶ The **equal-tailed** interval is $(q_{\alpha/2}, q_{1-\alpha/2})$.
- ▶ The **highest posterior density** interval searches for the smallest interval that contains the proper probability.

Credible intervals



In this beta/binomial example $Y = 8$, $N = 10$ and $a = b = 1$.

Hypothesis testing

- ▶ **Hypothesis tests** are conducted by simply computing the posterior probability of each hypothesis.
- ▶ Say the null hypothesis is $\mathcal{H}_0 : \theta \leq 0.5$ and the alternative is $\mathcal{H}_1 : \theta > 0.5$.
- ▶ The posterior probability of the null hypothesis is

$$\text{Prob}(\theta < 0.5 | Y) = \int_0^{0.5} p(\theta | Y) d\theta.$$

- ▶ We reject the null if its probability is small.
- ▶ In a Bayesian analysis we can say “Given the data and prior, the probability that the null hypothesis is true is 0.02”.
- ▶ This is analogous to but different than the **p-value**.

Hypothesis testing

```
> # Data
> Y <- 8; n <- 10
> # The posterior is  $\theta|Y \sim \text{Beta}(A, B)$ 
> A <- Y+1; B <- n-Y+1
> # Posterior mean
> A/(A+B)
[1] 0.75
> # Posterior standard deviation
> sqrt(A*B/((A+B)^2*(A+B+1)))
[1] 0.1200961
> # Posterior 95% credible interval
> qbeta(c(0.025, 0.975), A, B)
[1] 0.4822441 0.9397823
> # Posterior probability that  $\theta < 0.5$ 
> pbeta(0.5, A, B)
[1] 0.03271484
```

Monte Carlo approximations

- ▶ **Monte Carlo (MC) sampling** is a useful tool for summarizing a posterior.
- ▶ For univariate cases, is it not particularly useful. But, in harder problems, it is the best approach available.
- ▶ In MC sampling, we draw S samples from the posterior:

$$\theta^{(1)}, \dots, \theta^{(S)} \sim p(\theta | Y),$$

and use these samples to approximate the posterior.

- ▶ For example, the posterior mean and variance are approximated by the sample mean and variance of the $\theta^{(s)}$, $s = 1, \dots, S$.

Monte Carlo approximations

- ▶ MC sampling facilitates studying **transformations** of parameters.
- ▶ For example, the odds corresponding to θ are $\gamma = \theta/(1 - \theta)$.
- ▶ How to approximate the posterior mean and variance of γ ?
- ▶ We simply transform each draw to the odds

$$\gamma^{(1)} = \frac{\theta^{(1)}}{1 - \theta^{(1)}}, \dots, \gamma^{(S)} = \frac{\theta^{(S)}}{1 - \theta^{(S)}},$$

and use these draws to approximate γ 's posterior.

Monte Carlo approximations

```
> # Data
> Y <- 8; n <- 10
> # The posterior is  $\theta|Y \sim \text{Beta}(A, B)$ 
> A <- Y+1; B <- n-Y+1
> # MC sampling
> theta <- rbeta(100000, A, B)
> # Approximate the posterior mean and SD
> mean(theta); sd(theta)
[1] 0.749792
[1] 0.1201799
> # Transform to odds
> gamma <- theta/(1-theta)
> # Approximate the posterior mean and SD
> mean(gamma); sd(gamma)
[1] 4.483378
[1] 4.720541
```

Thank you!